

Object Oriented Programming 4 (3+1)

Course Logs

INSTITUTION	Department of Computer Science, Bahria University Karachi Campus, Karachi 73500.
PROGRAM (S) TO BE EVALUATED	BACHELOR OF SCIENCE (Information Technology)
Course Name	Object Oriented Programming 4 (3+1)
Catalog Number	CSC- 210
Instructor Name	Dr. Abdullah Ayub

Course learning outcomes/expected performance criteria:

Objectives: The course aims to provide command of the computer object-oriented programming language, is the most popular language worldwide. Most of the mobile android devices, and mobile games written in Java, allowing maximum control with minimal commands. A career in computer programming, it would be wise to start by learning the Java programming language.

Major Topics to be covered:

Evolution of Object-Oriented Programming (OOP), Object Oriented concepts and principles, problem solving in Object Oriented paradigm, OOP design process, classes, functions/methods, objects and encapsulation; constructors and destructors, operator and function/method overloading, association, aggregation, composition, generalization, inheritance and its types, derived classes, function/method overriding, abstract and concrete classes, virtual functions, polymorphism, exception handling.

Course Description This course provides introduction & overview of Computer Programming, Principles of Object-oriented Programming, Overview of OOP, OOP design process, classes, functions/methods, objects and encapsulation, polymorphism, inheritance & abstraction.

Course Outcomes:

At the completion of this course, students should be able to do the following:

- Being able to work with abstract concepts and being able to do technical analysis.
- Able Figures out the process of designing, writing, testing, debugging/troubleshooting and maintaining the source code of computer programs.
- The code modification of an existing source or something completely new program to develop.
- Writing source code including knowledge of the application domain, specialized algorithms and formal logic.
- Know the main building blocks of Object-Oriented Programming.

Course Title: Object Oriented Programming

Offered to: BS (IT)

Course Codes: CSC- 210

Credit Hours: (4) 3+1

Pre-requisite: N/A

Instructor: * Dr. Abdullah Ayub

Office Location: * Department of Computer Science

Email Address: abdullahayub.bukc@bahria.edu.pk

Reference:

1. An Introduction to Object-Oriented Programming with Java, C. Thomas Wu (2010). 5th Edition. McGraw-Hill. ISBN: 9780073523309
2. Java: How to Programme, 5/e, Deitel and Deitel, Prentice Hall, 0131016210/ 0131202367 International Edition.

Assessment:

Quizzes/Assignments / Attendance / Class Participation	30 Marks
Mid-term	20 Marks
Final Examination	50 Marks
Total	100 Marks

Course Outline (Weekly-Wise):

Week	Lecture	Topic
1	1	Evolution of Object-Oriented Programming (OOP)
	2	Evolution of Object-Oriented Programming (OOP)
2	3	Object Oriented Concepts and Principles
	4	Object Oriented Concepts and Principles
3	5	Problem Solving & Testing in Object Oriented paradigm
	6	OOP Design Process
4	7	Classes
	8	Classes
5	9	Functions/Methods
	10	Functions/Methods
6	11	Assignment/Quiz/Class Activity
	12	Operators in OOP
7	13	Construction/Destruction
	14	Construction/Destruction (Programming Perspectives)
Week #8 MID TERM EXAM		

9	17	Abstraction
	18	Abstraction
10	19	Encapsulation
	20	Encapsulation
11	21	Inheritance
	22	Inheritance
12	23	Polymorphism
	24	Polymorphism
13	25	Quiz/Assignment/Class Activity
	26	Overloading/Overriding
14	27	Exceptional Handling
	28	Virtual Function
15	29	Final Presentation
	30	Final Presentation
	Week #16 (FINAL EXAM)	

COURSE EVALUATION:

Type of Assessment	Percentage (%)	Remark
Quiz	10	10 Marks
Assignments	20	2 Assignments/Presentation (10x2=20 Marks)
Midterm Exam	20	20 Marks
Final Term Exam	50	50 Marks

COURSE OBJECTIVE:

The course aims to provide command of the computer object-oriented programming language, is the most popular language worldwide. Most of the mobile android devices, and mobile games written in Java, allowing maximum control with minimal commands. A career in computer programming, it would be wise to start by learning the Java programming language.

COURSE LEARNING OUTCOMES (CLOs):

Upon completion of this course, students will be able to:

CLO #	CLO Statement
CLO 1	Being able to work with abstract concepts and being able to do technical analysis.
CLO 2	Able Figures out the process of designing, writing, testing, debugging/troubleshooting and maintaining the source code of computer programs. The code modification of an existing source or something completely new program to develop.
CLO 3	Writing source code including knowledge of the application domain, specialized algorithms and formal logic.
CLO 4	Know the main building blocks of Object-Oriented Programming.

RECOMMENDED BOOKS:

Textbook(s)	Title	An Introduction to Object-Oriented Programming with Java, C. Thomas Wu (2010). 5th Edition. McGraw-Hill. ISBN: 9780073523309
	Author	Thomas Wu (2010)
	Publisher	McGraw-Hill.
Ref. Book(s)	Title	Java: How to Programme, 5/e, Deitel and Deitel, Prentice Hall, 0131016210/ 0131202367 International Edition.
	Author	Deitel and Deitel
	Publisher	Prentice Hall

GRADING SCHEME:

Marks	Grade	Marks	Grade
87-100	A	50 – 59	D
80 – 86	B+	< 50	F
72 – 79	B	Withdraw	W
66 – 71	C+	Incomplete	I
60 – 65	C		

COURSE POLICY:

Homeworks (Assignments)	- Homeworks are due at the beginning of class, - Late homeworks will not be accepted, - All works have to be done independently, - Students handing in similar homeworks will receive a grade of 0 (ZERO) and face possible disciplinary actions.
Makeup Exams	In accordance with university regulations, i.e. students should bring a valid excuse authenticated through valid channels immediately within first week after the exam. Otherwise, they will be considered absent and no makeup exam for them.
Lab/Tutorials	All works done in Lab/Tutorials must be individuals.
Attendance	- Students are expected to attend all classes - Attendance is compulsory and will be taken regularly. - Attendance for less than 80% of the lectures will result in students being barred from taking the Final Exam.
Code of Conduct	The assignments, quizzes and exams need to be done individually. Copying of another student's work or code, even if changes are subsequently made, is inappropriate, and such work or code will not be accepted . The University has very clear guidelines for academic misconduct, and they will be enforced in this class.
Cell Phone	Cell phones are restricted during class. Cell phones must be turned off during the lecture. If your cell phone rings during class, you may be asked to leave.

**It is acknowledged that the objectives, synopsis of the course and distributions of examination marks will not be changed in the framework of the course as certified.*

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Dr. Abdullah Ayub

Course Supervisor

Date:

Validated by Head of Department

Date: